

Chapter 1

MHF4U

State the degree and the leading coefficient of each polynomial

Polynomial	Degree	Leading Coefficient
$y = 2x^3 + 3x - 1$		
$y = 5x - 6$		
$y = x^3 - 2x^2 - 5x^4 + 3$		
$y = -3x^5 + 2x^3 - x - 1$		
$y = 21 - 2x + 4x^2 - 6x^3$		

Match each function to its end behavior

$$y = 3x^7$$

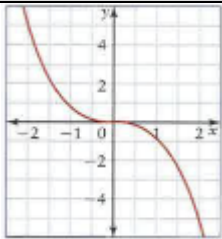
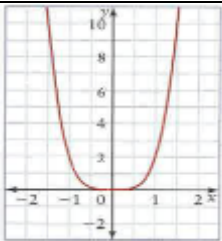
$$y = -\frac{1}{2}x^3$$

$$y = 2x^4$$

$$y = -0.25x^6$$

End Behaviour	Functions
Q3 to Q1	
Q2 to Q4	
Q2 to Q1	
Q3 to Q4	

Complete the following table

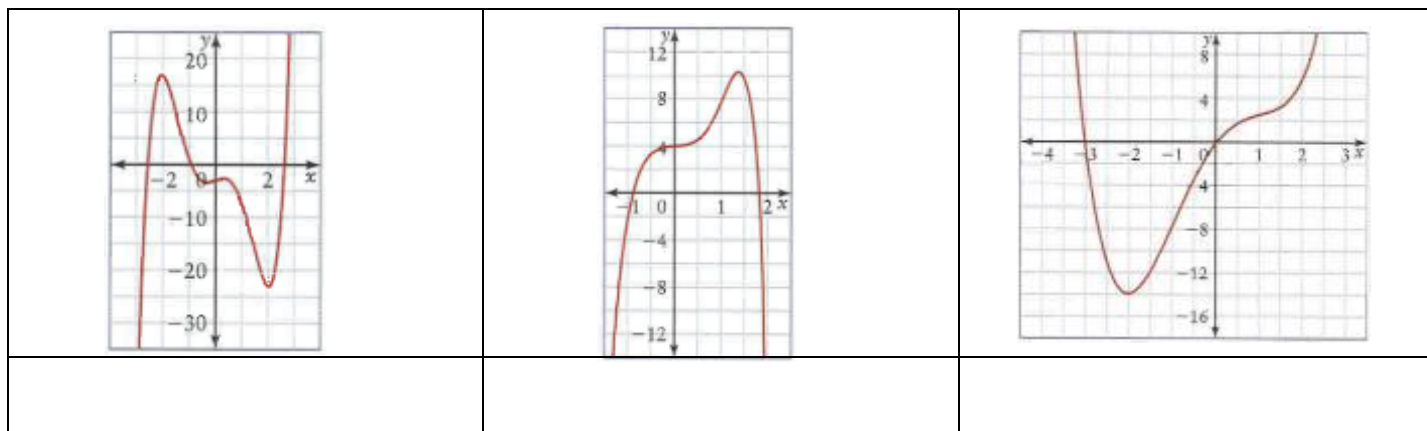
Graph of Function	Even or Odd Degree?	Sign of Leading Coefficient	Domain and Range	Symmetry	End Behaviour
					
					

Use end behaviours, turning points, and zeros to match each equation with the most likely graph. Write the letter of the equation beneath the graph.

A) $g(x) = 0.5x^4 - 3x^2 + 5x$

B) $h(x) = x^5 - 7x^3 + 2x - 3$

C) $p(x) = -x^6 + 5x^3 + 4$



Complete the following table

Equation	Degree	Sign of Leading Coefficient	End Behaviour	Possible number of turning points	Possible number of x-intercepts
$f(x) = 6x^3 + 2x$					
$g(x) = -20x^6 - 5x^3 + x^2 - 17$					
$p(x) = 22x^4 - 4x^3 + 3x^2 - 2x + 2$					
$h(x) = -x^5 + x^4 - x^3 + x^2 - x + 1$					

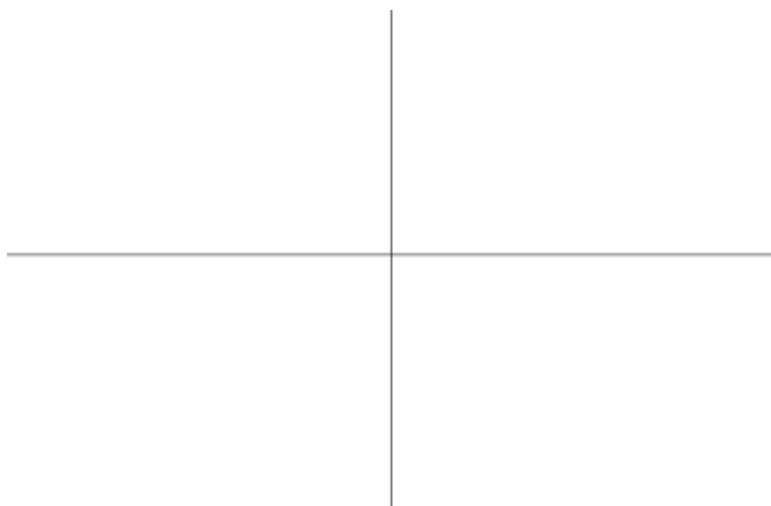
Complete the following table

Graph	Sign of Leading Coefficient	Even or Odd Degree?	End Behaviour	Symmetry	Number of turning points	Number of x-intercepts	Least Possible Degree

For each function, complete the chart and sketch a possible graph of the function labelling key points.

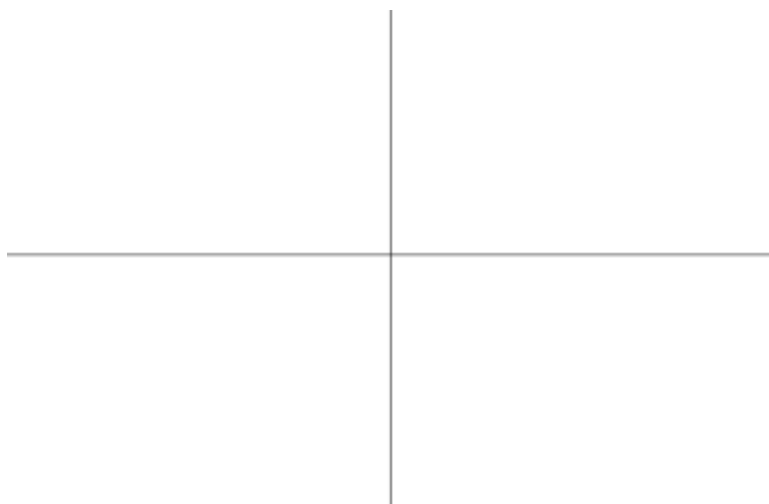
a) $f(x) = (x + 1)(x - 3)(x + 2)$

Degree	Leading Coefficient	End Behaviour	x -intercepts	y -intercept



b) $g(x) = -x(x + 1)(x + 2)^2$

Degree	Leading Coefficient	End Behaviour	x -intercepts	y -intercept



Write an equation for the function that results from the given transformations.

a) The function $f(x) = x^4$ is compressed vertically by a factor of $\frac{3}{5}$, stretched horizontally by a factor of 2, reflected horizontally in the y -axis, and translated 1 unit up and 4 units to the left.

b) The function $f(x) = x^3$ is compressed horizontally by a factor of $\frac{1}{4}$, stretched vertically by a factor of 5, reflected vertically in the x -axis, and translated 2 units to the left and 7 units up.

Identify the a , k , d and c values and explain what transformation is occurring to the parent function for $g(x) = 2[-4(x + 7)]^4 - 1$

For the following questions, use the key points of the parent function to perform transformations. Graph the parent and transformed function. Write the equation of the transformed function.

a) $f(x) = x^3$ $g(x) = \frac{1}{2}f(x + 2) - 4$

